Predicting Limit Order Fill Probability Across 5, 30 and 60 Seconds

Technical report on a synthetic AAPL limit order book dataset

Executive summary

This study estimates whether a resting limit order receives its first fill within 5, 30 or 60 seconds. It uses the supplied chronological train, validation and test split, constructs features only from information available before observation, and evaluates separate probability models at each horizon.

0.170	0.600	0.805
5-second test PR-AUC	30-second test PR-AUC	60-second test PR-AUC

- All three learned model families improved on the empirical fill-rate baseline on validation data.
- The frozen test policy used Logistic Regression at 5 seconds and calibrated Random Forests at 30 and 60 seconds.
- Queue position dominates model explanations; recent aggressive trade flow becomes more important at longer horizons.
- Features from the 50 pre-submission snapshots did not produce a stable gain beyond the static order book and recent-flow feature set.

The 5-second task remained difficult because only about one in ten eligible orders filled. Results should be interpreted as evidence about this deterministic synthetic dataset, not real AAPL execution.

1. Problem, data and evaluation protocol

For each eligible active order and horizon H, the target is the probability that the first post-observation fill occurs by t0 + H. Partial and complete first fills both count as positive outcomes.

Data package

Source	Contents	Role
Orders workbook	10,000 orders; 72 order-level fields	Static t0 state, labels and supplied split
History JSON	50 snapshots per order; 500,000 snapshots	Pre-submission dynamics and feature set D

Chronology and target eligibility

Stage	Orders	Trading days	Use
Train	7,000	First 14	Fit models and preprocessing
Validation	1,500	Next 3	Select candidate, calibration and threshold
Test	1,500	Final 3	One frozen evaluation

Horizon	Eligible rows	Overall positive rate	Interpretation
5 seconds	9,722	10.6%	Rare and strongly imbalanced fills
30 seconds	8,808	41.9%	Moderate class balance
60 seconds	8,252	63.2%	Fills form the majority class

Leakage controls

- The supplied data_split defined the chronology. No random row split was used.
- Eligibility filtering preceded modeling; blank targets were never converted to non-fills.
- Outcome fields, post-t0 events, identifiers and metadata were excluded from predictors.
- The test labels remained untouched until the model, calibration method and threshold were frozen.

Random seed 42 was used throughout. Final confidence intervals used 1,000 bootstrap resamples.

2. Models, feature ablation and validation findings

The experiment compared an empirical fill-rate baseline, Logistic Regression, a small Decision Tree and Random Forest. Four nested feature sets isolated studied-order variables (A), contemporaneous LOB state (B), recent flow (C) and 50-snapshot history (D).

Best validation PR-AUC within each model family

Horizon	Empirical	Logistic	Tree	Random Forest
5s	0.102	0.248 (B)	0.192 (C)	0.235 (C)
30s	0.396	0.599 (B)	0.540 (C)	0.598 (D)
60s	0.614	0.786 (C)	0.729 (B)	0.789 (C)

RQ1 - Model performance

Every learned family improved on the empirical baseline at every horizon. Logistic Regression led at 5 and 30 seconds in the common-settings comparison, while Random Forest produced the highest 60-second PR-AUC. The result does not support the claim that nonlinearity always dominates a well-specified linear model.

RQ2 - Value of LOB history

Horizon	Best non-history candidate	PR-AUC	Full-history candidate	PR-AUC
5s	B_logistic	0.247	D_logistic	0.198
30s	C_random_forest	0.604	D_random_forest	0.601
60s	C_random_forest	0.798	D_random_forest	0.796

Feature set D did not improve the selected validation candidate at any horizon. The history features remain methodologically valid, but their incremental signal was too small or redundant with static LOB and recent-flow variables in this dataset.

Model selection rule

For each candidate, calibration was chosen by validation Brier score. The final candidate was then selected by validation PR-AUC, and its classification threshold maximized validation F1. This separates ranking, probability quality and the operational decision threshold.

3. Frozen test evaluation

Horizon	Frozen candidate	PR-AUC	ROC-AUC	Brier	ECE	F1	Recall
5s	Logistic B	0.170	0.616	0.102	0.014	0.158	0.122
30s	RF C	0.600	0.650	0.230	0.035	0.611	0.746
60s	RF C	0.805	0.690	0.205	0.028	0.789	0.985

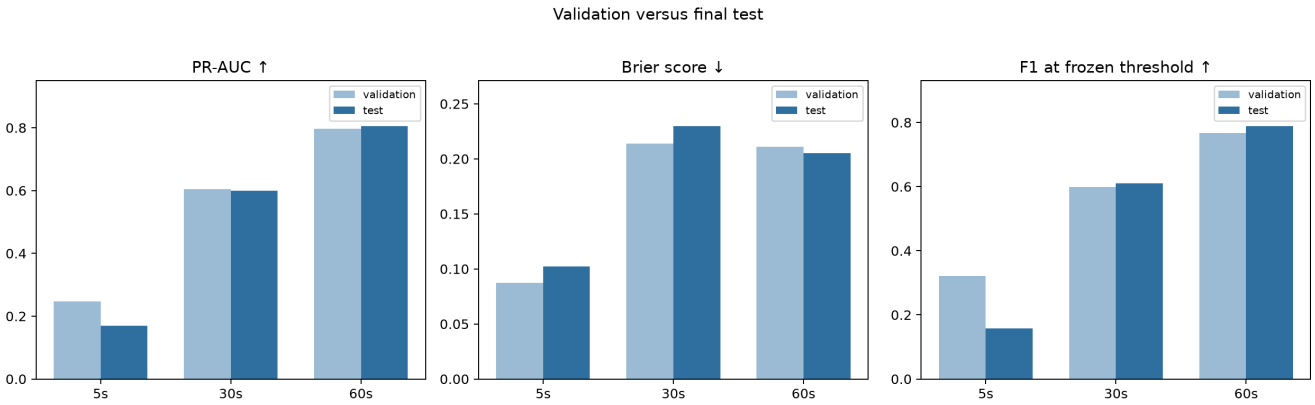


Figure 1. Validation and frozen test performance. Source: notebook 06 and results/validation_test_comparison.csv.

Operational interpretation

- At 5 seconds, PR-AUC fell from 0.247 on validation to 0.170 on test. Only 21 of 172 fills were detected at the frozen F1 threshold, so this horizon should be treated cautiously.
- At 30 seconds, PR-AUC remained stable at 0.600. Recall reached 0.746, with 435 true positives and 407 false positives.
- At 60 seconds, PR-AUC reached 0.805 and recall 0.985. The low threshold classified 96.4% of eligible orders as fills, leaving only 33 true negatives.

RQ4 - Probability quality

The best-ranking and best-calibrated candidates were close but not always identical. At 60 seconds, the D Random Forest had a marginally lower validation Brier score than C, while C had the higher PR-AUC. The frozen test ECE values of 0.014, 0.035 and 0.028 indicate small average calibration gaps, but threshold behavior still matters operationally.

4. Global model behavior

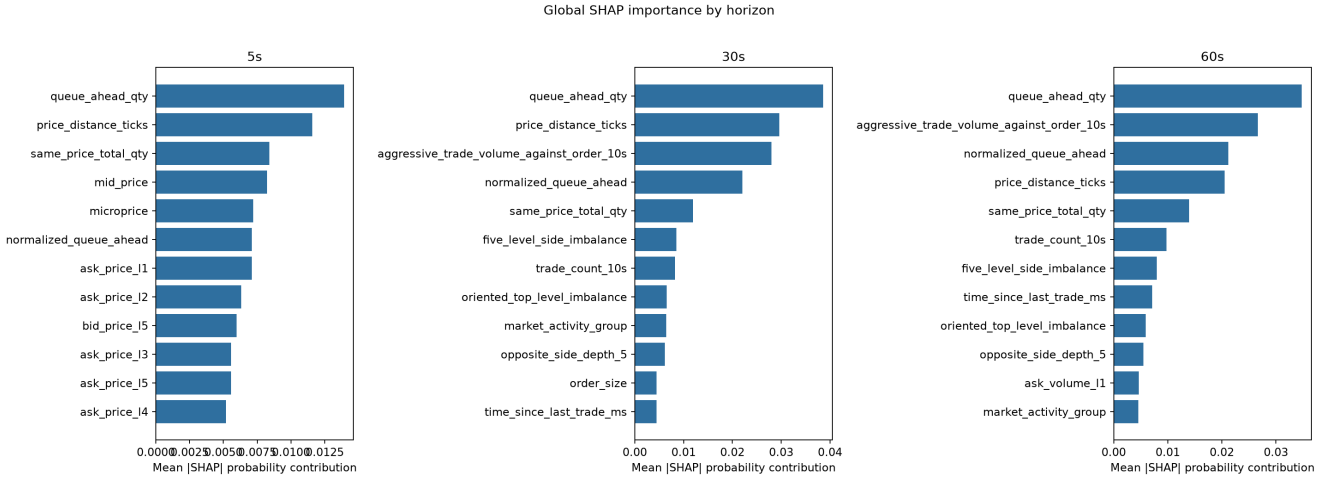


Figure 2. Global mean absolute SHAP contributions for Random Forest feature set C across horizons. Source: notebook 08.

Queue and price priority

Queue ahead quantity ranked first at all horizons. Price distance and normalized queue position also remained prominent. These variables represent how much displayed liquidity must be cleared and how competitive the order price is.

The role of recent flow

Aggressive trade volume against the order became the second-ranked SHAP feature at 60 seconds and third at 30 seconds. Recent executions provide more time to accumulate into a fill signal at the longer horizons.

Decision-tree market paths at 30 seconds

Case	Order	Readable path	Leaf fill probability
High Probability	ORD008229	queue_ahead_qty <= 250.5 AND aggressive_trade_volume_against_order_10s > 248	0.740
Low Probability	ORD008395	queue_ahead_qty > 250.5 AND price_distance_ticks > 1.5	0.143
Uncertain	ORD007710	queue_ahead_qty <= 250.5 AND aggressive_trade_volume_against_order_10s <= 248	0.496

Impurity and permutation importance disagreed strongly at 5 and 30 seconds: their top-10 Jaccard values were 0.176 and 0.250. Agreement rose to 0.667 at 60 seconds. Correlated LOB features can divide importance and make any single ranking unstable.

5. Local explanations and stability

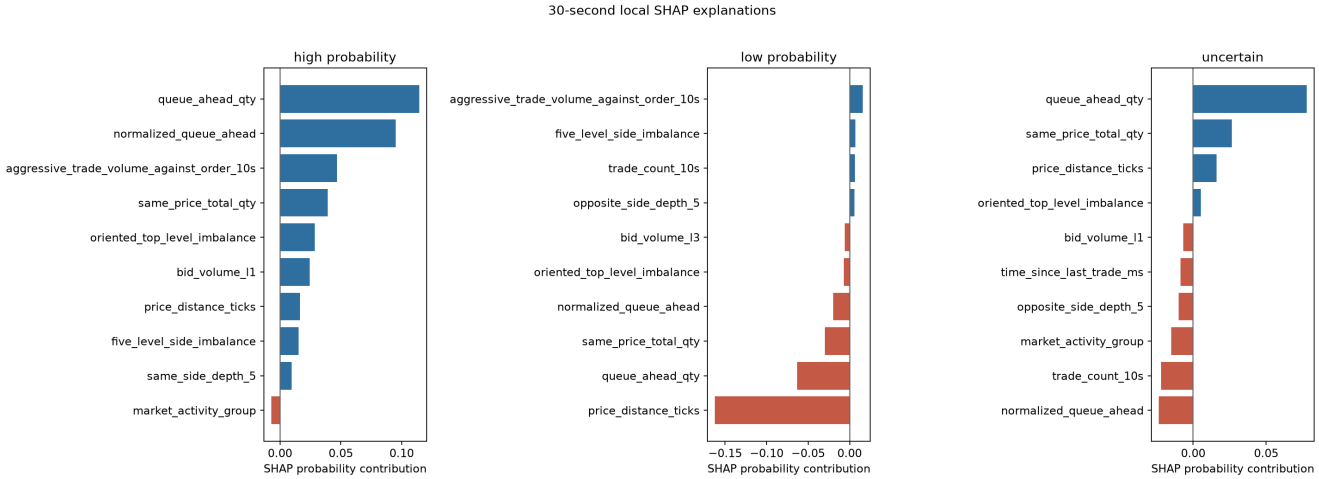


Figure 3. SHAP explanations for the same high, low and uncertain 30-second cases used in the LIME analysis. Source: notebook 08.

Uncertain case ORD007710

The 30-second Random Forest assigned probability 0.500 and the order did fill. Low queue-ahead quantity raised the prediction, while normalized queue position, trade count and market-activity group pulled it downward. The near-balanced contributions explain why the probability remained close to 0.5.

LIME stability over five random seeds

Case	Mean top-feature Jaccard	Minimum Jaccard	Mean sign agreement
High Probability	0.806	0.667	1.000
Low Probability	0.776	0.667	0.975
Uncertain	0.821	0.667	1.000

Case	SHAP/LIME top-10 overlap	Jaccard	Shared-sign agreement
High Probability	7	0.538	1.000
Low Probability	8	0.667	1.000
Uncertain	8	0.667	1.000

LIME was reasonably stable under the specified sampling setup, and shared SHAP/LIME features had consistent signs. The methods still differed in which lower-ranked correlated variables entered the top ten. Neither method establishes causality.

RQ3 - Horizon

The 5-second target is more imbalanced and offers less time for queue depletion. At 30 and 60 seconds, aggressive trade flow and cumulative queue conditions become more informative, while the positive class becomes much more common.

6. Research-question answers and conclusion

Question	Answer
RQ1: Model performance	Yes. All learned model families beat the empirical validation baseline. Logistic Regression remained competitive; Random Forest led at 60 seconds.
RQ2: LOB history	No stable incremental gain. Feature set D did not exceed the best non-history candidate at any horizon.
RQ3: Horizon	The 5-second task is rare-event prediction. Queue priority matters throughout; aggressive trade flow gains importance at longer horizons.
RQ4: Probability quality	Ranking and calibration were related but not identical. The 60-second C model ranked best while D had a marginally lower validation Brier score.
RQ5: Stability	Performance was broadly similar across sides and days at 30/60 seconds, with meaningful subgroup variation. The 5-second model was less stable because fills were rare.

RQ5 evidence

On test data, 30-second PR-AUC was 0.616 for buys and 0.585 for sells; daily values ranged from 0.582 to 0.633. At 60 seconds, buy/sell PR-AUC was 0.823/0.787 and daily values ranged from 0.791 to 0.822. These gaps do not invalidate the model, but they argue against presenting one aggregate metric as universal performance.

Limitations

- The data are deterministic and synthetic, cover one symbol, and do not represent live Nasdaq execution.
- The study estimates association and predictive probability, not causal effects of changing market variables.
- Correlated price and depth variables complicate feature attribution.
- The frozen F1 thresholds create different operating regimes across horizons, especially the high predicted-fill rate at 60 seconds.
- Fixed-horizon classification discards some censored observations; survival analysis remains an optional extension.

Conclusion

A chronology-respecting model can rank fill probabilities meaningfully, especially at 30 and 60 seconds. Most of the useful signal comes from the studied order's queue position, its price competitiveness and recent aggressive flow. Additional pre-submission history did not improve the selected models, so a simpler static-plus-flow specification is preferable for this dataset.

Reproducibility and sources

All reported values come from committed notebooks 01-08 and the CSV/JSON files under results/. The project brief is AI_Order_Execution_Project.pdf. The raw sources are AI_Order_Execution.xlsx and the paired 50-snapshot JSON history. Final policy commit provenance and bootstrap settings are recorded in results/final_test_summary.json.